Experiment No. 2

Signal Operations and Transformations

OBJECTIVE 1:

To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

Procedure

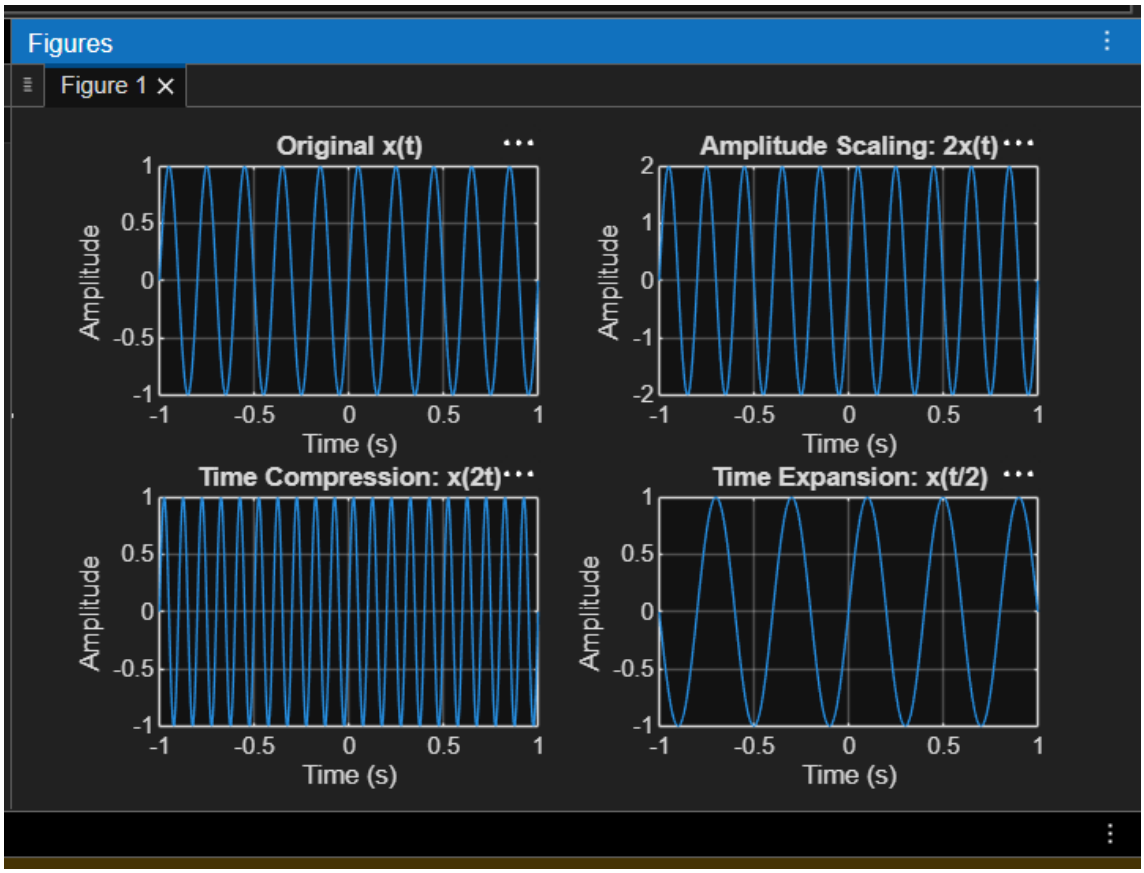
1. Generate the base signal x(t) = sin(2π5t).
2. Apply amplitude scaling using A = 2.
3. Apply time compression using x(2t).
4. Apply time expansion using x(t/2).
5. Plot the original and transformed signals using common axes.
6. Measure the amplitude and period of the transformed signals.
7. Compare the measured results with the theoretical scaling relationships.

MATLAB Coding

```
% Experiment 2(i): Amplitude and Time Scaling
clc; clear; close all;
t = -1:0.001:1;
x = sin(2*pi*5*t);
yA = 2*x;
yC = sin(2*pi*5*(2*t));
yE = sin(2*pi*5*(t/2));
figure;
subplot(2,2,1); plot(t,x); grid on;
title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,2); plot(t,yA); grid on;
```

```
title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,3); plot(t,yC); grid on;
title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,4); plot(t,yE); grid on;
title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```

OUTPUT:



RESULT:

• Periodic Signal: The sinusoidal signal repeats itself at regular intervals, making it periodic.

• Aperiodic Signal: The exponentially decaying signal does not repeat, showcasing that it is aperiodic.

This test case illustrates the key differences between periodic and aperiodic signals by generating and visually analysing both types using MATLAB

OBJECTIVE 2:

To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

Procedure

1. Define the sample index n.

2. Enter x[n] and y[n] with aligned indices.

3. Compute the sum sequence.

4. Compute the product sequence using pointwise multiplication.

5. Plot the input and resulting sequences using stem plots.

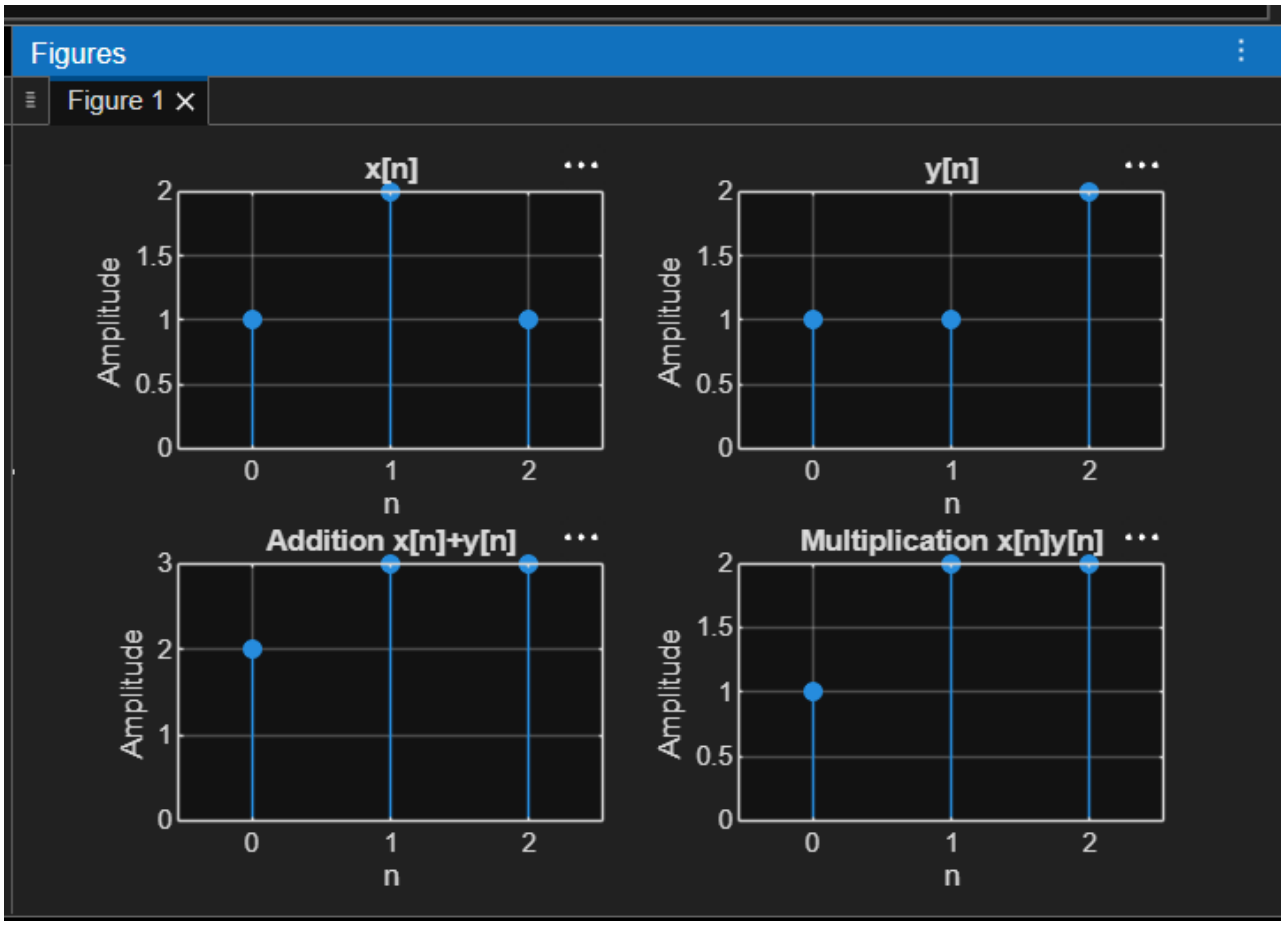
6. Compare the calculated samples with manual/theoretical results.

MATLAB Coding

```
% Experiment 2(ii): Addition and Multiplication of Two Sequences
clc; clear; close all;
n = 0:2;
x = [1 2 1];
y = [1 1 2];
zSum = x + y;
zProd = x .* y;
figure;
subplot(2,2,1); stem(n,x,'filled'); grid on;
title('x[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,2); stem(n,y,'filled'); grid on;
title('y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,3); stem(n,zSum,'filled'); grid on;
title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,4); stem(n,zProd,'filled'); grid on;
```

title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');

OUTPUT:



RESULT:

Scaling affects amplitude or time axis accordingly.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and growth or decay. Explains the	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	